# (5) Improved retrieval with Rag Fine-tuning.

## (1) Adapt and fine-tune Rag models

(V1) → Adapt Rag models ~~into~~ using Financial Domain Data

→ Adapting a retrieval augmented generation model using financial domain data involves fine-tuning both the rag components with specialised data set.

→ For ex → consider a financial advisory form aiming to enhance its automated support system.

- For this they will collect a data set of frequently asked questions, financial reports and articles on topics such as investment, market trends and financial regulations.

V2 → Task specific fine-tuning using BERT

→ BERT → stands for Bidirectional Encoder representations from transformers.

→ It is a natural language processing model developed by google. ~~stands~~

→ Task-specific fine tuning using bert for (rag) model involves adapting the model to perform a particular task with high accuracy.

→ Fais → Facebook AI library developed by Facebook for similarity search and clustering of dense vectors.

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V3 Domain Adaptation Fine-tuning using GPT →

→ Domain Adaptation fine-tuning of a RAG model using GPT, involves customising the model to work effectively within a specific domain by leveraging domain specific data.
→ This domain specific adaptation ensures the model can provide accurate and reliable medical advice.

V4 → Multi task Fine-tuning using T5

→ Multitask fine tuning using T5 model involves training the model to perform multiple related tasks such as question answering and summarization simultaneously
→ T5 model developed by Google is a versatile text to text transformer model that can handle various NLP tasks,

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